Quiver Representations

[Schiffler] [Etingof et. al. Ch. 6]

Def 93:

a) A <u>quiver</u> $Q = (Q_0, Q_1, s, t)$ is a directed graph w/

Q_0: vertices $\quad s: Q_1 \to Q_0$ "source"

Q_1: arrows $\quad t: Q_1 \to Q_0$ "target"

$$s(\alpha) \xrightarrow{\alpha} t(\alpha)$$

b) A <u>repn</u> of Q is an assignment of:

- to each vertex w, a v.s. V_w
- to each arrow $v \xrightarrow{\alpha} w$, a linear map $V_\alpha : V_v \to V_w$

(we will restrict to finite quivers and f.d. $\mathbb{C}$-v.s.)

c) A <u>morphism</u> $V \to W$ of Q repns is a collection of maps

$$\varphi_v : V_v \to W_v, \quad v \in Q_0,$$

that commute w/ the linear maps:

$$\begin{tikzcd} V_v \arrow[r, "V_\alpha"] \arrow[d, "\varphi_v"'] \arrow[dr, phantom, "\circlearrowright"] & V_w \arrow[d, "\varphi_w"] \\ W_v \arrow[r, "W_\alpha"'] & W_w \end{tikzcd} \qquad \forall \; v \xrightarrow{\alpha} w$$

Isomorphism, subrepn, quotient repn., direct sum, are defined similarly as for gps., Lie algebras, etc.

d) A Q-repn V is <u>simple</u> (or <u>irreducible</u>) if its only subrepns of V and $\{0\}$. V is <u>indecomposible</u> if

$V_v = \{0\}, V_\alpha = 0$

$V \cong V' \oplus V'' \Rightarrow V'$ or $V'' = 0$.

By the Krull-Schmidt Theorem, every f.d. quiver repn can be written as a direct sum of indecomposibles, unique up to isom. / ordering.

Examples:

a) Vertex: $Q = \bullet$

repns: $\mathbb{C}^n$

only one indecomposible: $\mathbb{C}$ (also simple)

b) Arrow: $Q = \bullet \to \bullet$

repns: $\mathbb{C}^m \xrightarrow{A} \mathbb{C}^n$ $\qquad A \in \mathrm{Mat}_{n,m}$

Choosing bases of $\mathbb{C}^m, \mathbb{C}^n$, can write

$$A = \underbrace{I_r}_{r := \operatorname{rank} A} \oplus 0 \qquad \begin{array}{ccc} \mathbb{C}^r & \xrightarrow{I} & \mathbb{C}^r \\ \oplus & & \oplus \\ \underbrace{\mathbb{C}^{m-r}}_{\ker A} & & \underbrace{\mathbb{C}^{n-r}}_{\operatorname{coker} A} \end{array}$$

Indecomposibles: $\underset{S(1)}{\mathbb{C} \to 0}$, $\underset{S(2)}{0 \to \mathbb{C}}$, $\underset{P(1)}{\mathbb{C} \xrightarrow{1} \mathbb{C}}$

not simple!

Morphisms:

$$\begin{tikzcd}
S(2) \arrow[d, hook] & 0 \arrow[r] \arrow[d] & \mathbb{C} \arrow[d, "1"] \\
P(1) \arrow[d, two heads] & \mathbb{C} \arrow[r, "1"] \arrow[d, "1"'] & \mathbb{C} \arrow[d] \\
S(1) & \mathbb{C} \arrow[r] & 0
\end{tikzcd}$$

c) Loop: $Q = \bigcirc$ repns: $\mathbb{C}^m \circlearrowleft A$ $A \in \mathrm{End}(\mathbb{C}^m)$

indecomposibles $\longleftrightarrow$ Jordan blocks

$$J_{\lambda,n} = \underbrace{\begin{bmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix}}_{n} \quad \begin{matrix} \lambda \in \mathbb{C} \\ n \geq 1 \end{matrix}$$

d) Kronecker quiver $\bullet \rightrightarrows \bullet$ Repns: $V \underset{B}{\overset{A}{\rightrightarrows}} W$

Indecomps. w/ $\dim V = \dim W = 1$ are

$\mathbb{C} \underset{b}{\overset{a}{\rightrightarrows}} \mathbb{C}$ a, b not both 0 up to mutual scalar i.e. pts $[a:b]$ in $\mathbb{P}^1(\mathbb{C})$

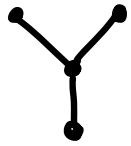
e) A_n Dynkin diag., any orientation:

e.g. $\bullet \to \bullet \to \bullet \leftarrow \bullet \to \bullet \to \bullet \leftarrow \bullet \leftarrow \bullet$

Indecomposibles correspond to intervals:

$$V(i,j): \quad \overset{1}{0} - 0 - \cdots - 0 - \overset{i}{\mathbb{C}} \overset{1}{-} \mathbb{C} - \cdots \overset{1}{-} \overset{j}{\mathbb{C}} - 0 - \cdots \overset{n}{0}$$

f) D_4 Dynkin diagram



Simples:

$$\begin{tikzcd} \mathbb{C} \arrow[r] & 0 & 0 \arrow[l] \\ & 0 \arrow[u] & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & 0 & \mathbb{C} \arrow[l] \\ & 0 \arrow[u] & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & 0 & 0 \arrow[l] \\ & \mathbb{C} \arrow[u] & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & 0 \arrow[l] \\ & 0 \arrow[u] & \end{tikzcd}$$

Other indecomposibles:

$$\begin{tikzcd} \mathbb{C} \arrow[r, "1"] & \mathbb{C} & 0 \arrow[l] \\ & 0 \arrow[u] & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & \mathbb{C} \arrow[l, "1"'] \\ & 0 \arrow[u] & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & 0 \arrow[l] \\ & \mathbb{C} \arrow[u, "1"] & \end{tikzcd}$$

$$\begin{tikzcd} \mathbb{C} \arrow[r, "1"] & \mathbb{C} & \mathbb{C} \arrow[l, "1"'] \\ & 0 \arrow[u] & \end{tikzcd} \qquad \begin{tikzcd} \mathbb{C} \arrow[r, "1"] & \mathbb{C} & 0 \arrow[l] \\ & \mathbb{C} \arrow[u, "1"] & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & \mathbb{C} \arrow[l, "1"'] \\ & \mathbb{C} \arrow[u, "1"] & \end{tikzcd}$$

$$\begin{tikzcd} \mathbb{C} \arrow[r, "1"] & \mathbb{C} & \mathbb{C} \arrow[l, "1"'] \\ & \mathbb{C} \arrow[u, "1"] & \end{tikzcd} \qquad \begin{tikzcd} \mathbb{C} \arrow[r, "\begin{bmatrix}1\\0\end{bmatrix}"] & \mathbb{C}^2 & \mathbb{C} \arrow[l, "\begin{bmatrix}0\\1\end{bmatrix}"'] \\ & \mathbb{C} \arrow[u, "\begin{bmatrix}1\\1\end{bmatrix}"'] & \end{tikzcd}$$

Three subspaces:

$$\begin{tikzcd} U \arrow[dr, hook] & & V \arrow[dl, hook] \\ & U+V+W & \\ & W \arrow[u, hook] & \end{tikzcd}$$

$$\dim(U+V+W) = \dim U + \dim V + \dim W - \dim(U\cap V) - \dim(U\cap W) - \dim(V\cap W) + \dim(U\cap V\cap W) + \#\left(\begin{tikzcd} \mathbb{C} \arrow[r] & \mathbb{C}^2 & \mathbb{C} \arrow[l] \\ & \mathbb{C} \arrow[u] & \end{tikzcd}\right)$$

(mathoverflow/23478: "Most common mathematical false belief")

Thm 94 (Gabriel's Theorem): Q has finitely-many indecomposibles if and only if it is a union of orientations of Dynkin diagrams of type A_n, D_n, or E_n. In this case, indecomposibles are in bijection with positive roots.

A_n: $V(i,j) \leftrightarrow e_i - e_j \quad (i<j)$

D_4:

$$\begin{tikzcd} \mathbb{C} \arrow[r] & 0 & 0 \arrow[l] \\ & 0 \arrow[u] & \\ & \alpha_1 & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & 0 & \mathbb{C} \arrow[l] \\ & 0 \arrow[u] & \\ & \alpha_3 & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & 0 & 0 \arrow[l] \\ & \mathbb{C} \arrow[u] & \\ & \alpha_4 & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & 0 \arrow[l] \\ & 0 \arrow[u] & \\ & \alpha_2 & \end{tikzcd}$$

$$\begin{tikzcd} \mathbb{C} \arrow[r] & \mathbb{C} & 0 \arrow[l] \\ & 0 \arrow[u] & \\ & \alpha_1+\alpha_2 & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & \mathbb{C} \arrow[l] \\ & 0 \arrow[u] & \\ & \alpha_2+\alpha_3 & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & 0 \arrow[l] \\ & \mathbb{C} \arrow[u] & \\ & \alpha_2+\alpha_4 & \end{tikzcd}$$

$$\begin{tikzcd} \mathbb{C} \arrow[r] & \mathbb{C} & \mathbb{C} \arrow[l] \\ & 0 \arrow[u] & \\ & \alpha_1+\alpha_2+\alpha_3 & \end{tikzcd} \qquad \begin{tikzcd} \mathbb{C} \arrow[r] & \mathbb{C} & 0 \arrow[l] \\ & \mathbb{C} \arrow[u] & \\ & \alpha_1+\alpha_2+\alpha_4 & \end{tikzcd} \qquad \begin{tikzcd} 0 \arrow[r] & \mathbb{C} & \mathbb{C} \arrow[l] \\ & \mathbb{C} \arrow[u] & \\ & \alpha_2+\alpha_3+\alpha_4 & \end{tikzcd}$$

$$\begin{tikzcd} \mathbb{C} \arrow[r] & \mathbb{C} & \mathbb{C} \arrow[l] \\ & \mathbb{C} \arrow[u] & \\ & \alpha_1+\alpha_2+\alpha_3+\alpha_4 & \end{tikzcd} \qquad \begin{tikzcd} \mathbb{C} \arrow[r] & \mathbb{C}^2 & \mathbb{C} \arrow[l] \\ & \mathbb{C} \arrow[u] & \\ & \alpha_1+2\alpha_2+\alpha_3+\alpha_4 & \end{tikzcd}$$

Def 95: The <u>path algebra</u> of a quiver Q is the $\mathbb{C}$-alg. $\mathbb{C}Q$ spanned by all paths in Q, including the "lazy paths" $\{e_v, v \in Q_0\}$, with relations

$$p \cdot q = \{\underbrace{p \circ q}_{\text{concatenation}} \text{ if } q \text{ ends where } p \text{ begins, else } 0\}$$

The category of f.d. left $\mathbb{C}Q$-modules is equiv. to the category of f.d. Q-repns. via

$$M \mapsto V \quad \text{where } V_v = e_v M$$

$$\begin{tikzcd} v \arrow[r, "\alpha"] & w \end{tikzcd} \qquad \begin{aligned} V_\alpha : e_v M &\to e_w M \\ e_v m &\mapsto \alpha m \end{aligned}$$

$$V \mapsto M = \bigoplus_{v \in Q_0} V_v \quad \mathbb{C}Q\text{-action by the maps } V_\alpha.$$

Ex: $Q = \begin{tikzcd} 1 \arrow[r, "\alpha"] & 2 \arrow[r, "\beta"] & 3 \end{tikzcd}$

$$\mathbb{C}Q = \operatorname{span}\{e_1, e_2, e_3, \alpha, \beta, \beta\alpha\} \cong \begin{bmatrix} \mathbb{C} & 0 & 0 \\ \mathbb{C} & \mathbb{C} & 0 \\ \mathbb{C} & \mathbb{C} & \mathbb{C} \end{bmatrix} \qquad \begin{bmatrix} e_1 & & \\ \alpha & e_2 & \\ \beta\alpha & \beta & e_3 \end{bmatrix}$$